

Resampling Techniques

Problem 9: Confidence Interval Comparison

Using a real dataset (e.g., housing prices), estimate mean θ .
Construct:

- Normal CI
- Bootstrap- t CI
- Bootstrap Percentile CI
- BC_a CI

Task 9.1: Normal and Bootstrap- t CI

- 1 Compute classical Normal CI.
- 2 Compute Bootstrap- t CI.
- 3 Compare symmetry and width.

Question:

- Why does studentization help?

Task 9.2: Percentile and BC_a CI

- 1 Compute Percentile CI.
- 2 Compute BC_a CI.
- 3 Compare location and skewness adjustment.

Discussion:

- When do Percentile and BC_a differ significantly?
- Which is transformation-respecting?

Deliverable: Problem 9

- Table of all four intervals
- Comparison of CI lengths
- Comment on skewness and bias correction

Problem 10: Hypothesis Testing

We test:

$$H_0 : \mu_1 = \mu_2$$

using:

- Bootstrap test
- Permutation test
- Studentized bootstrap test

Example dataset:

```
from sklearn.datasets import load_wine
```

Task 10.1: Bootstrap Test

- 1 Compute observed statistic:

$$t = \bar{X} - \bar{Y}$$

- 2 Generate bootstrap samples from pooled data.
- 3 Compute achieved significance level (ASL).

Task 10.2: Permutation Test

- 1 Randomly permute combined sample.
- 2 Compute statistic for each permutation.
- 3 Estimate p-value.

Discussion:

- Why is permutation exact under exchangeability?

Task 10.3: Studentized Bootstrap Test

- 1 Use studentized statistic:

$$\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\hat{\sigma}_1^2}{m} + \frac{\hat{\sigma}_2^2}{n}}}$$

- 2 Bootstrap under H_0 .
- 3 Compare p-values.

Deliverable: Problem 10

- Table of p-values (Bootstrap / Permutation / Studentized)
- Interpretation of differences
- Discussion on robustness and assumptions